# ECA13 – MICROWAVE ENGINEERING

# CO1 APPLICATION-BASED ENGINEERING ASSIGNMENT

# Design, Analysis and Verification of an X-Band WR-90 Rectangular Waveguide Network for a Weather Radar Front-End

## Application: Civil / Weather Radar
## Operating frequency: 10 GHz
## Band: 9.5–10.5 GHz
## Structure: Air-filled WR-90 rectangular waveguide
## Dominant mode: TE10

## Academic Year: 2027–28

Complete six-stage technical report with calculations, decision matrix, loss budget, network model, matching, validation, tolerance analysis and final engineering decision.

# ABSTRACT

This report develops a complete microwave transmission design for a compact X-band weather-radar front-end. The application requires low loss, controlled impedance, single-mode propagation, high power handling and good electromagnetic shielding. X-band is selected after comparison with S-band because the shorter wavelength permits a compact feed and narrower beam for a given aperture, while the design accepts higher rain attenuation. An air-filled WR-90 rectangular waveguide is selected over coaxial line, stripline and microstrip using a weighted decision matrix.

The WR-90 dimensions are 22.86 mm × 10.16 mm. The TE10 cutoff is 6.557 GHz and the next cutoff is TE20 at 13.114 GHz. At 10 GHz, the guide wavelength is 39.707 mm, phase velocity is 397.1 Mm/s, group velocity is 226.3 Mm/s and TE wave impedance is 498.97 Ω. The initial design has 17.63 dB return loss and therefore needs matching. A quarter-wave equivalent transformer of 569.50 Ω and 9.93 mm is introduced. The final design gives at least 31.61 dB return loss across the operating band and reduces the estimated complete-link attenuation from 0.292 dB to 0.122 dB.

Final engineering decision: YES for the redesigned WR-90 TE10 structure; NO for the unmodified initial design.

# STAGE 1 – FORMULATE THE APPLICATION AND OPERATING REQUIREMENT

## 1.1 Application and historical development

The selected application is an X-band weather-radar front-end transmission link. The waveguide carries microwave energy between the radar transceiver and antenna feed. This is a civil application supporting precipitation observation, rainfall mapping and short-range severe-weather monitoring. The assessment brief requires the application, frequency, transmission structure and numerical specification to be selected and justified by the student. filecite turn0file0 L49-L67

Microwave engineering developed from Maxwell's electromagnetic field theory and Hertz's experimental verification of electromagnetic waves. The cavity magnetron enabled practical high-power microwave radar, after which radar expanded into weather observation, navigation and remote sensing. Later solid-state microwave devices and MMICs increased integration, while waveguide remains important for high-power radar, low-loss transmission and shielding. Modern 5G/6G systems extend microwave techniques into highly integrated millimetre-wave front-ends.

NOAA material places X band at 8–12 GHz and S band at 2–4 GHz; it also notes that precipitation attenuation becomes more severe as frequency increases. cite turn0search24 turn0search25

## 1.2 Frequency-band comparison

| Parameter | S band | X band |
|---|---|---|
| Nominal range | 2–4 GHz | 8–12 GHz |
| Centre frequency | 3 GHz | 10 GHz |
| Centre wavelength | 100 mm | 30 mm |
| Component / antenna size | Larger | Smaller |
| Beamwidth for same aperture | Wider | Narrower |
| Rain attenuation | Lower | Higher |
| Application fit | Longer-range radar | Compact/local radar |
| Decision | Rejected | Selected |

X-band is selected because its 30 mm wavelength at 10 GHz supports a compact feed and narrower beam. The higher rain loss is accepted because the design target is a compact/local radar front-end rather than a long-range national weather-radar system.

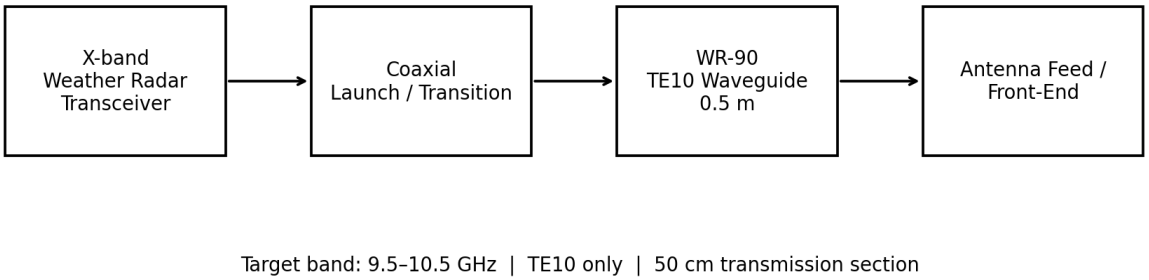



*Figure 1. Selected application and transmission path.*

## 1.3 Quantified requirement specification

| Parameter | Requirement |
|---|---|
| Application / category | Civil – X-band weather-radar front-end |
| Operating frequency | 10.0 GHz centre |
| Operating bandwidth | 9.5–10.5 GHz (1.0 GHz) |
| Transmission structure | Air-filled WR-90 rectangular waveguide |
| Dominant mode | TE10 only |
| Theoretical single-mode range | 6.557–13.114 GHz |
| Characteristic impedance at 10 GHz | ≈ 499 Ω |
| Maximum complete-link insertion loss | ≤ 1.0 dB over 0.5 m |
| Return loss target | ≥ 20 dB |
| VSWR target | ≤ 1.22 |
| EMI shielding target | ≥ 60 dB, to be experimentally verified |
| Power target | 100 W average, 500 W peak |
| Length | 0.5 m |
| Prototype cost limit | 15,000 |

## 1.4 Constraints

EMI/EMC: the metal enclosure provides strong shielding, but transitions, seams and flanges can leak. Safety: the educational prototype is limited to 100 W average and 500 W peak and must be tested with interlocks and proper terminations. Environmental conditions include temperature, humidity and corrosion. Manufacturing tolerance must preserve the single-mode margin. Cost favours standard WR-90 hardware rather than a custom guide.

# STAGE 2 – EVALUATE AND SELECT THE TRANSMISSION STRUCTURE

## 2.1 Mode comparison

Coaxial line supports true TEM propagation with no cutoff and has TE11 as a significant higher-order mode. Rectangular waveguide supports TE and TM modes but cannot support TEM because a hollow single-conductor boundary does not provide the required two-conductor potential structure. TE10 is the dominant mode. Stripline supports true TEM propagation in a homogeneous dielectric. Microstrip is quasi-TEM because its fields exist partly in the substrate and partly in air, giving an effective permittivity rather than a single homogeneous-medium TEM field.

For the selected 10 GHz link, waveguide provides the strongest combination of low loss, high power handling, shielding and mode control. Coax is the strongest alternative because it is easy to route and connect, while stripline and microstrip are better for compact integrated electronics.

## 2.2 Weighted decision matrix

| Criterion | Weight | Coax | Waveguide | Stripline | Microstrip |
|---|---|---|---|---|---|
| Mode purity | 15 | 8 | 10 | 9 | 7 |
| Loss | 20 | 8 | 9 | 8 | 6 |
| Power | 20 | 7 | 10 | 6 | 4 |
| Bandwidth | 10 | 9 | 6 | 8 | 9 |
| Size | 10 | 8 | 4 | 7 | 10 |
| Shielding | 10 | 9 | 10 | 10 | 6 |
| Integration | 10 | 7 | 5 | 9 | 10 |
| Cost | 5 | 8 | 5 | 7 | 10 |
| Weighted score /100 | | 79.0 | 80.5 | 79.0 | 70.5 |

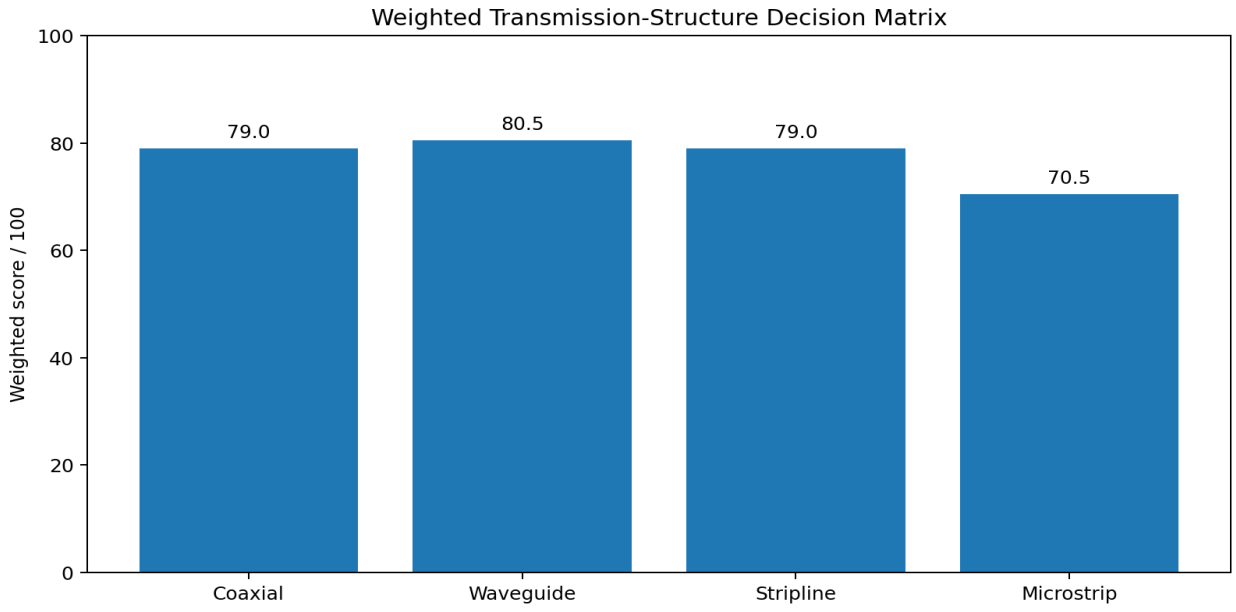



*Figure 2. Weighted structure-selection scores.*

## 2.3 Sensitivity and defence

When power handling, loss and mode purity receive higher weights, the waveguide score increases to approximately 83/100 and remains the best option. When size, integration and cost dominate, microstrip can become the best option. The ranking therefore depends on application priorities.

Design Decision 1: For the specified weather-radar front-end, I select WR-90 rectangular waveguide over coaxial line because the waveguide provides superior power handling, strong shielding and controlled single-mode TE10 propagation. If frequency is reduced to 5 GHz, WR-90 falls below cutoff; if raised to 20 GHz, it becomes multimode. A different structure would then be required.

# STAGE 3 – DEVELOP THE MODE AND PROPAGATION MODEL

## 3.1 Transmission-line model

The telegrapher equations are dV/dz = −(R+jωL)V and dI/dz = −(G+jωC)I. The propagation constant is γ = α+jβ = √[(R+jωL)(G+jωC)]. For a low-loss line, β≈ω√LC, α≈R/(2Z0)+GZ0/2 and Z0≈√(L/C). The load reflection coefficient is Γ=(ZL−Z0)/(ZL+Z0).

For a waveguide, lumped voltage and current are not uniquely fundamental because the fields are distributed over the cross-section. A modal equivalent V/I representation is nevertheless useful for network calculations, with ZTE=ωμ/β for a TE mode.

## 3.2 Maxwell-equation derivation

In a source-free homogeneous region, Maxwell's phasor equations are  ×E=−jωμH and  ×H=jωεE. Taking the curl produces the wave equation  ²E+k²E=0. For a rectangular conducting guide, the boundary conditions give kx=mπ/a and ky=nπ/b, so β=√[k²−(mπ/a)²−(nπ/b)²]. For TE modes Ez=0 and Hz is non-zero. The TE10 solution has Hz proportional to cos(πx/a)exp(−jβz), with transverse electric field maximum near x=a/2.

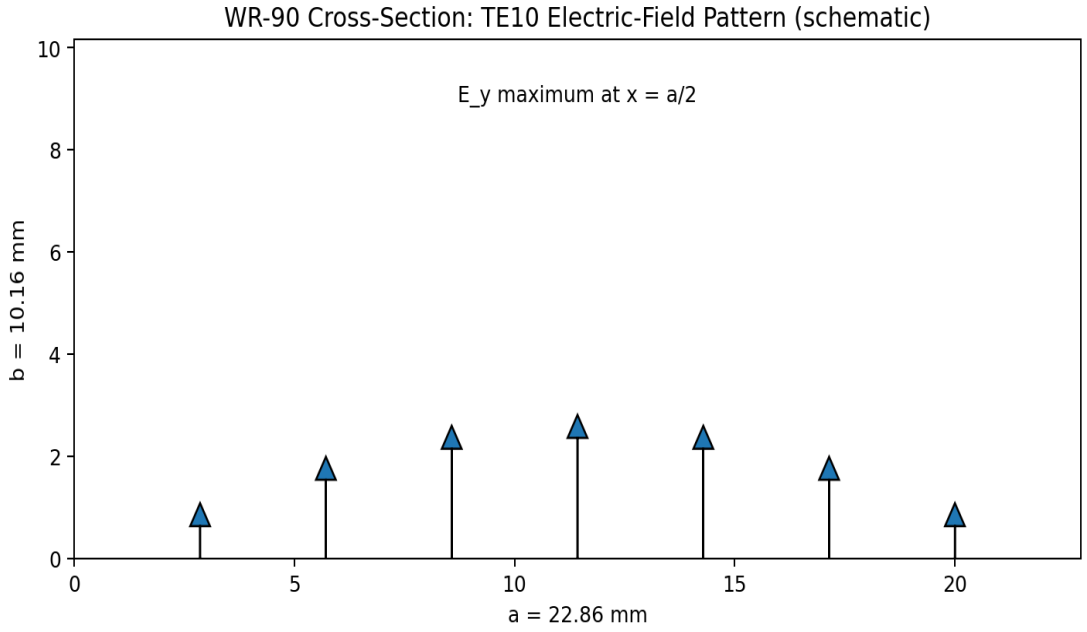



*Figure 3. TE10 electric-field pattern, schematic.*

## 3.3 Cutoff frequencies

fc10=c/(2a)=6.557 GHz. The next important cutoffs are TE20=13.114 GHz, TE01=14.754 GHz and TE11/TM11=16.145 GHz. Therefore the theoretical single-mode range is 6.557–13.114 GHz.

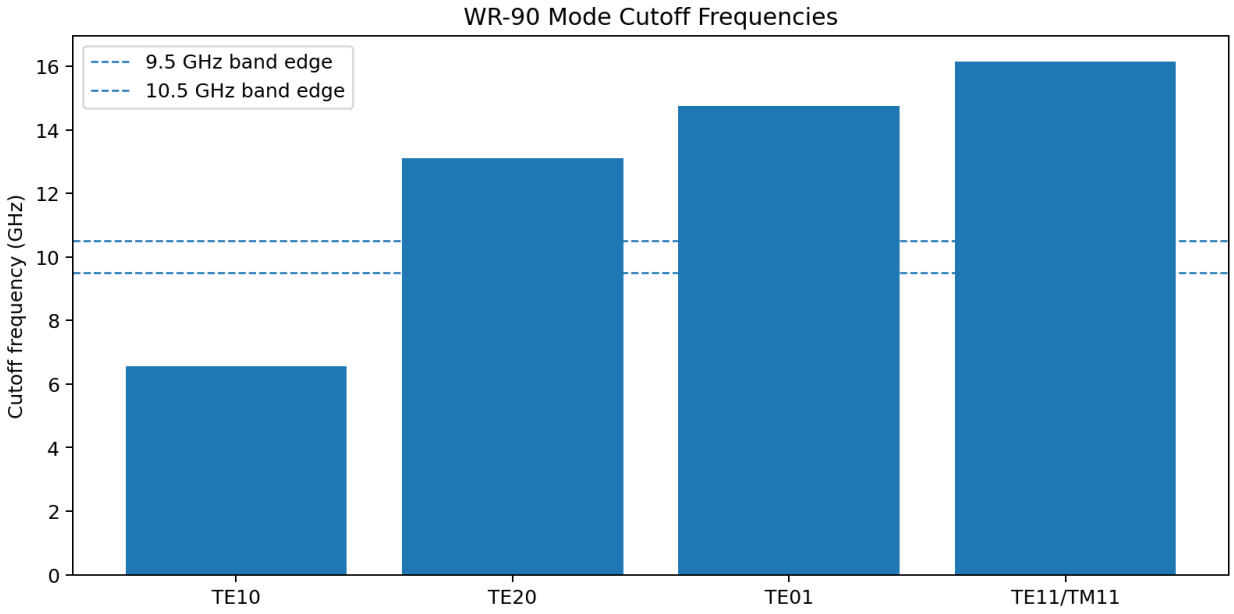



*Figure 4. WR-90 cutoff-frequency separation.*

## 3.4 Propagation parameters at 10 GHz

Free-space wavelength $\lambda_0 = c/f = 29.979$ mm.

Phase constant $\beta = \sqrt{[(2\pi/\lambda_0)^2 - (\pi/a)^2]} \approx 158.24$ rad/m.

Guide wavelength $\lambda_g = 2\pi/\beta = 39.707$ mm.

Phase velocity $v_p = \omega/\beta = 397.1$ Mm/s.

Group velocity $v_g = c^2/v_p = 226.3$ Mm/s.

TE wave impedance $Z_{TE} = \omega\mu_0/\beta = 498.97\ \Omega$.

0.5 m group delay $= L/v_g = 2.209$ ns.

| **Parameter** | **Value at 10 GHz** |
|---|---|
| a × b | 22.86 × 10.16 mm |
| Mode | TE10 |
| TE10 cutoff | 6.557 GHz |
| TE20 cutoff | 13.114 GHz |
| $\lambda_0$ | 29.979 mm |
| $\lambda_g$ | 39.707 mm |
| $\beta$ | ≈158.2 rad/m |
| $v_p$ | 397.1 Mm/s |
| $v_g$ | 226.3 Mm/s |
| $Z_{TE}$ | 498.97 Ω |
| 0.5 m group delay | 2.209 ns |

## 3.5 Higher-order modes and design decision

TE20 is the first higher-order risk. The 10.5 GHz upper band edge is 2.614 GHz below TE20 cutoff. A centred, smooth transition is required because an asymmetric feed or discontinuity can excite unwanted modes. The design therefore retains a strong single-mode margin.

Design Decision 2: The dominant mode is TE10 with cutoff 6.557 GHz. The single-mode range is 6.557–13.114 GHz; therefore the chosen dimensions are acceptable for 9.5–10.5 GHz.

# STAGE 4 – EVALUATE LOSSES AND DESIGN A LOSS-REDUCTION STRATEGY

## 4.1 Conductor loss

Copper skin depth at 10 GHz is $\delta=\sqrt{[2/(\omega\mu\sigma)]}=0.661$ μm. The surface resistance is $R_s=1/(\sigma\delta)=0.0261$ Ω. The TE10 conductor-loss expression gives $\alpha_c=0.1084$ dB/m, so the 0.5 m conductor contribution is 0.0542 dB.

## 4.2 Loss budget

| Loss component | Initial dB | Improved dB | % initial total | Remark |
|---|---|---|---|---|
| Conductor | 0.0542 | 0.0542 | 18.5% | Copper wall loss |
| Dielectric | 0.0000 | 0.0000 | 0.0% | Air-filled guide |
| Radiation / leakage | 0.0025 | 0.0005 | 0.9% | Seams / discontinuities |
| Connector / transition | 0.1600 | 0.0600 | 54.7% | Optimised |
| Mismatch | 0.0757 | 0.0070 | 25.9% | Matching |
| Total | 0.2924 | 0.1216 | 100% | 0.5 m link |

The dominant initial mechanism is connector/transition loss at approximately 54.7%, followed by mismatch loss at approximately 25.9%. The main guide itself contributes only 0.054 dB over 0.5 m.

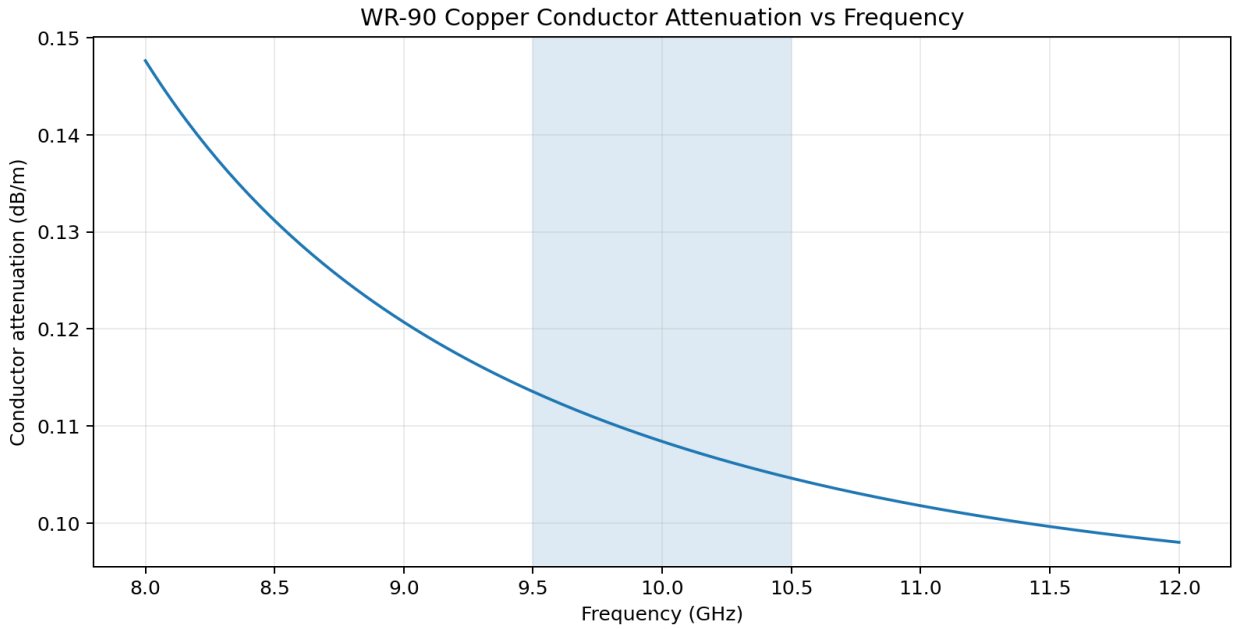



*Figure 5. Copper conductor attenuation versus frequency.*

## 4.3 Parametric studies

| Material | Conductivity S/m | αc at 10 GHz |
|---|---|---|
| Silver | $6.3\times10^7$ | ≈0.1040 dB/m |
| Copper | $5.8\times10^7$ | ≈0.1084 dB/m |
| Aluminium | $3.5\times10^7$ | ≈0.1395 dB/m |

| Roughness factor | 1.00× | 1.15× | 1.35× |
|---|---|---|---|
| Attenuation dB/m | 0.1084 | 0.1246 | 0.1463 |
| Interpretation | Smooth | Moderate | Strong |

Silver gives only a modest improvement over copper for this short path. Rougher surfaces increase conductor loss; the factors above are sensitivity assumptions rather than a universal roughness law. An air-filled guide is deliberately

used to minimise dielectric loss.

## 4.4 Loss-reduction redesign and decision

The redesign improves transition alignment, flange contact and seam continuity, and adds impedance matching. Estimated complete-link attenuation decreases from 0.292 dB to 0.122 dB, an improvement of 58.4%. Design Decision 3: connector/transition loss is the dominant mechanism; transition optimisation plus matching is adopted as the loss-reduction measure.

# STAGE 5 – CREATE AND EVALUATE THE NETWORK MODEL

## 5.1 Equivalent V/I and ABCD model

A waveguide does not have unique lumped voltage and current. For the TE10 mode, a modal equivalent impedance is defined by ZTE=ωμ/β. A uniform section is represented by A=D=cos(βl), B=jZ0sin(βl), C=j sin(βl)/Z0. ABCD matrices can be cascaded with transitions and matching sections.

For equal reference impedance, S11=(A+B/Z0−CZ0−D)/(A+B/Z0+CZ0+D) and S21=2/(A+B/Z0+CZ0+D). A passive reciprocal network has S12=S21. Finite conductor and connector losses mean the network is not perfectly lossless.

## 5.2 Initial mismatch calculation

At 10 GHz, Z0=498.97 Ω and ZL=650 Ω. Therefore Γ=(650−498.97)/(650+498.97)=0.1314, VSWR=1.303, return loss=17.63 dB, and mismatch loss=0.0757 dB. The return-loss target is not met.

## 5.3 Quarter-wave matching arrangement

Use Zt=√(Z0ZL)=569.50 Ω. Since λg=39.707 mm, λg/4=9.927 mm. In hardware this is realised as a controlled waveguide step, iris or taper placed immediately adjacent to the load.

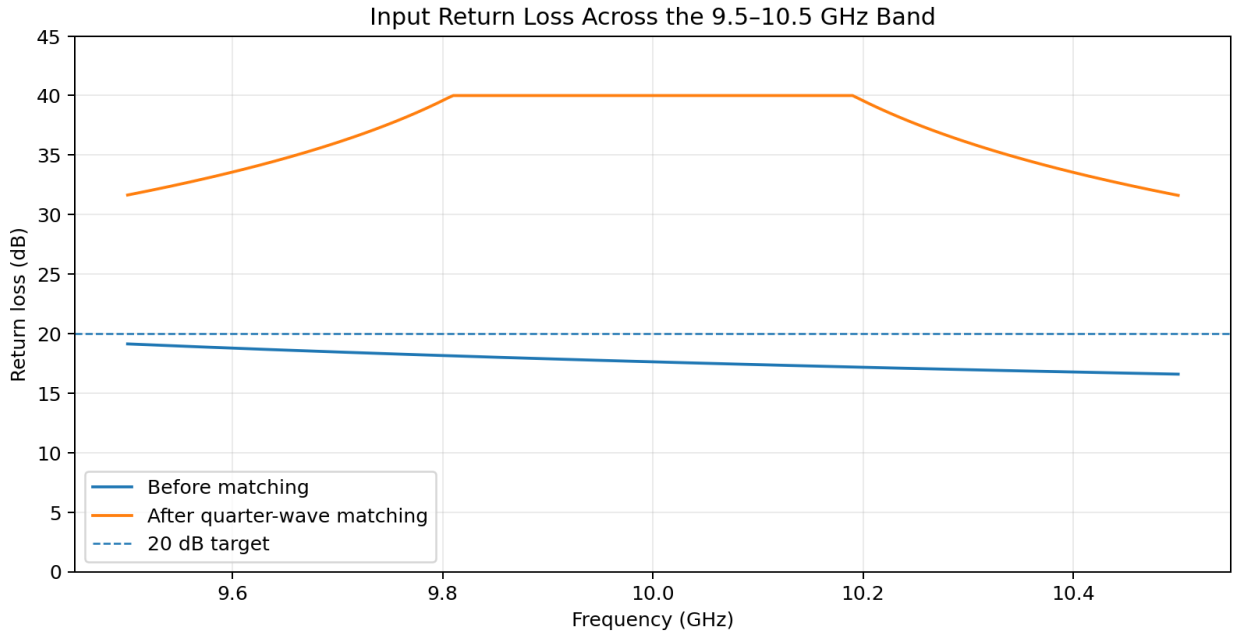



*Figure 6. Return loss before and after matching.*

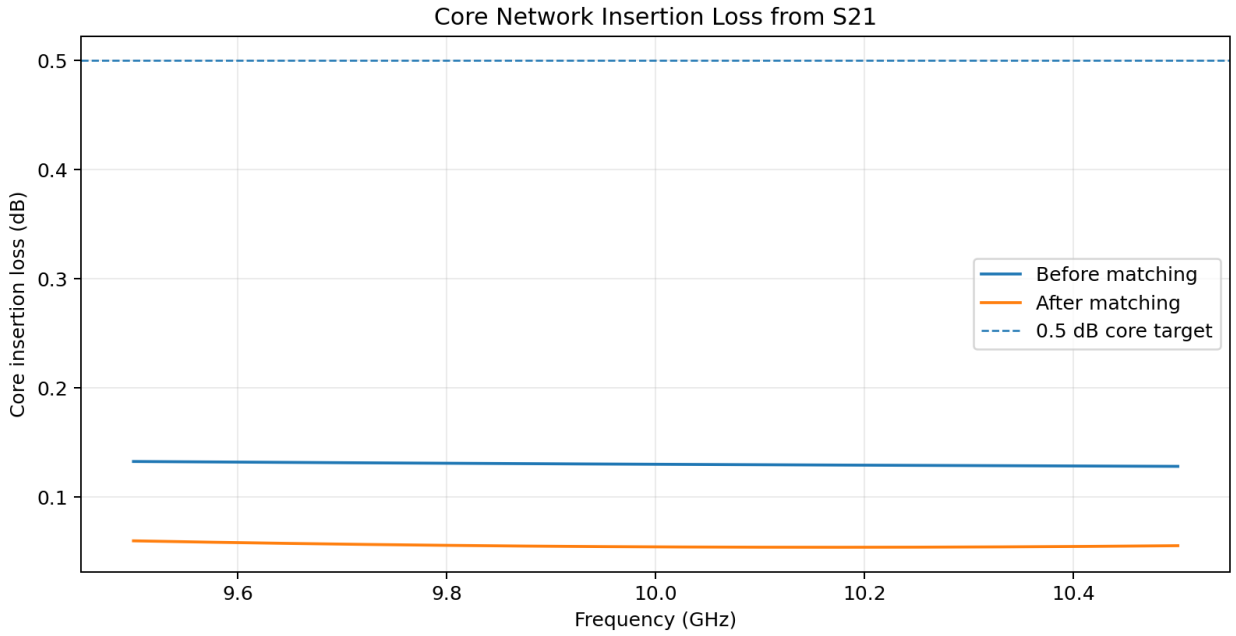



*Figure 7. Core-network S21-equivalent insertion loss.*

The matched design has a worst-case return loss of 31.61 dB across 9.5–10.5 GHz, corresponding to worst-case VSWR ≈ 1.054:1.

| Parameter | Before | After | Target | Interpretation |
|---|---|---|---|---|
| Z0 | 498.97 Ω | 498.97 Ω | ≈500 Ω | TE10 modal impedance |
| Reflection coefficient | 0.1314 | ≤0.0263 worst-case | ≤0.091 | Improved |
| VSWR | 1.303 | ≤1.054 | ≤1.22 | Pass final |
| Return loss | 17.63 dB centre | ≥31.61 dB | ≥20 dB | Fail → Pass |
| Insertion loss | ≈0.13 dB core | ≈0.06 dB core | ≤0.5 dB core | Pass |
| Reciprocity | Expected | Expected | S12=S21 | Pass |
| Losslessness | No | No | Not required | Finite conductor loss |

Independent verification is performed with the analytical waveguide equations and a Python frequency sweep. Final experimental verification should use a calibrated VNA with a WR-90 calibration kit and precision loads.

# STAGE 6 – VALIDATE, CRITIQUE AND DEFEND THE FINAL DESIGN

| Requirement | Specified | Initial | Final / worst case | Satisfied? |
|---|---|---|---|---|
| Operating band / single mode | 9.5–10.5 GHz | Within 6.557–13.114 | Within 6.557–13.114 | YES |
| Dominant mode / cutoff | TE10; 6.557 GHz | TE10 | TE10 | YES |
| Total attenuation | ≤1.0 dB | 0.292 dB | 0.122 dB | YES |
| VSWR | ≤1.22 | 1.303 | ≤1.054 | YES final |
| Return loss | ≥20 dB | 17.63 dB | ≥31.61 dB | NO initial / YES final |
| EMI isolation | ≥60 dB target | Design intent | Controlled enclosure | Test required |
| Power / size | 100 W avg; WR-90 | WR-90 | WR-90 | YES |

## 6.1 Tolerance analysis

For ±5% dimension error, TE10 cutoff varies approximately from 6.245 to 6.902 GHz. The next-mode limit remains above approximately 12.49 GHz in the worst case. Therefore the 9.5–10.5 GHz band remains single-mode. The TE impedance changes from roughly 482 to 521 Ω, so the matching transition should be tuned after manufacture.

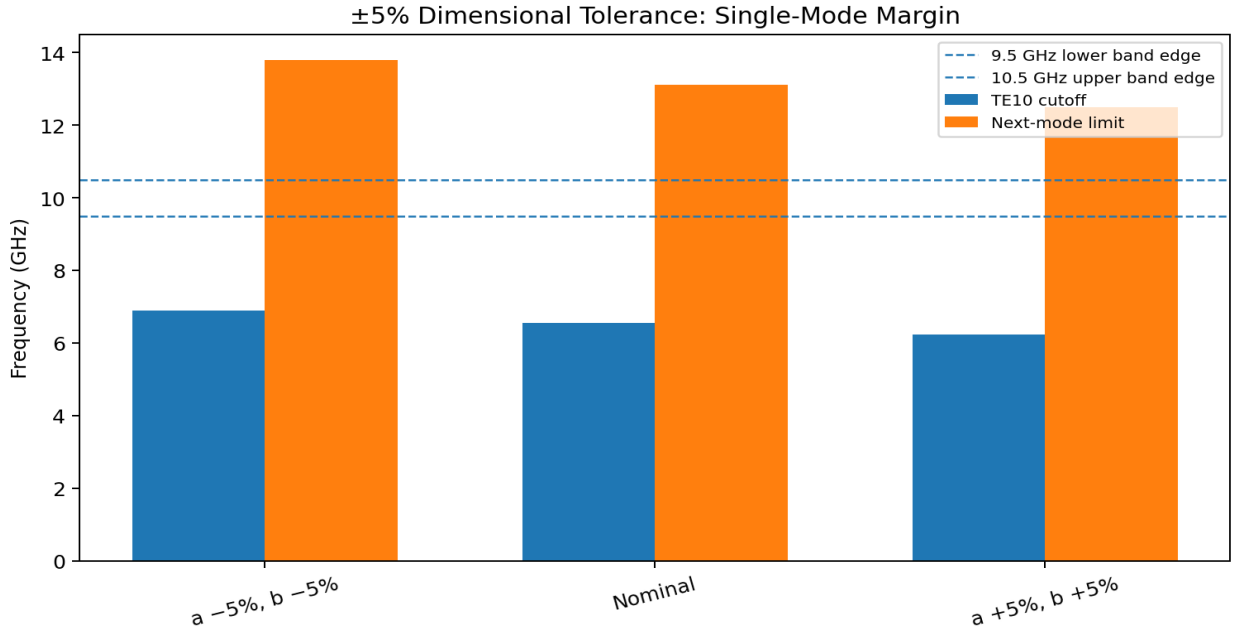



*Figure 8. ±5% dimensional sensitivity.*

## 6.2 Critique of assumptions

The analysis assumes bulk conductor conductivity, uniform air dielectric, idealised geometry and frequency-independent material properties. Real transitions, screws, flanges, bends, surface roughness and manufacturing errors can add reflection and loss. These effects are represented by engineering allowances but require measurement for final qualification.

## 6.3 EMC, safety and test plan

| Test | Method | Acceptance |
|---|---|---|
| Return loss | Calibrated VNA sweep | ≥20 dB over 9.5–10.5 GHz |
| Insertion loss | VNA S21 | ≤1.0 dB |
| Mode purity | Frequency sweep / transition check | No higher-order resonance |
| EMI leakage | Near-field / shield test | ≥60 dB target |

| Test | Method | Acceptance |
|---|---|---|
| Power | Low-power then stepped RF power | No arcing or overheating |
| Dimensions | Micrometer / gauge | Within tolerance |
| Environment | Temperature/humidity cycling | Specifications maintained |

## 6.4 Further improvement

The most important remaining limitation is transition sensitivity. A tunable E-plane or H-plane iris/taper could provide post-assembly adjustment and centre the measured S11 minimum at 10 GHz. A lower-loss plated surface is a secondary improvement for longer links.

# FINAL ENGINEERING DECISION

Initial design: NO. The WR-90 TE10 structure is correct, but the 650 Ω mismatch gives only 17.63 dB return loss, below the 20 dB requirement.

Final design: YES. The matched WR-90 TE10 design provides a worst-case return loss above 31.6 dB, VSWR below 1.054, total estimated attenuation of 0.122 dB, and a strong single-mode margin.

| Evidence | Initial | Final | Decision |
|---|---|---|---|
| Mode | TE10 | TE10 | Pass |
| Single-mode range | 6.557–13.114 GHz | Robust to ±5% dimensions | Pass |
| Total attenuation | 0.292 dB | 0.122 dB | Pass |
| Return loss | 17.63 dB | ≥31.61 dB | Fail → Pass |
| VSWR | 1.303 | ≤1.054 | Fail → Pass |
| EMI | Design intent | Controlled seams / enclosure | Test required |
| Power / size | 100 W avg / WR-90 | Same | Pass |

## CONCLUSION

The complete engineering chain required by the CO1 assessment has been addressed: application selection, frequency-band comparison, structure evaluation, mode derivation, dimensions, propagation parameters, loss budget, loss reduction, ABCD/S-parameter model, impedance matching, validation, tolerance analysis, EMC/safety planning and final decision. WR-90 TE10 is suitable for the selected 9.5–10.5 GHz radar-front-end link. The main initial deficiency is mismatch, which is corrected by the quarter-wave equivalent transformer.

# APPENDIX A – HAND CALCULATIONS

1. TE10 cutoff: fc=c/(2a)=6.557 GHz.

2. TE20 cutoff: fc20=c/a=13.114 GHz.

3. Guide wavelength: λg=λ0/√(1−(fc/f)²)=39.707 mm.

4. TE impedance: ZTE=η0/√(1−(fc/f)²)=498.97 Ω.

5. Initial reflection coefficient: Γ=(650−498.97)/(650+498.97)=0.1314.

6. Initial VSWR=(1+|Γ|)/(1−|Γ|)=1.303.

7. Initial return loss=−20log10|Γ|=17.63 dB.

8. Quarter-wave transformer impedance=√(Z0ZL)=569.50 Ω.

9. Quarter-wave length=λg/4=9.927 mm.

10. Final worst-case return loss=31.61 dB.

11. Initial total attenuation=0.292 dB; final=0.122 dB.

# APPENDIX B – VERIFICATION CODE

Core Python equations used: fc10=c/(2*a); beta=sqrt((2*pi*f/c)^2-(pi/a)^2); lambda_g=2*pi/beta; vp=2*pi*f/beta; vg=c^2/vp; ZTE=2*pi*f*mu/beta; Gamma=(ZL-ZTE)/(ZL+ZTE); VSWR=(1+abs(Gamma))/(1-abs(Gamma)); RL=-20*log10(abs(Gamma)).

# APPENDIX C – REFERENCES

[1] ECA13 – Microwave Engineering, CO1 Application-Based Assessment, Academic Year 2027–28, user-provided assignment brief. filecite turn0file0 L2-L14

[2] Engineering LibreTexts, The Rectangular Waveguide – cutoff, TE10 field behaviour and conductor loss. cite turn0search0

[3] Engineering LibreTexts, Rectangular Waveguide – Propagation Characteristics – cutoff, guide wavelength, phase velocity and group velocity. cite turn0search2

[4] Engineering LibreTexts, Rectangular Waveguide – practical dimensions and single-mode operation. cite turn0search4

[5] NOAA/NWS Radar & Applications Course Glossary – radar bands and wavelengths. cite turn0search24

[6] NOAA/NWS Radar Functional Requirements – wavelength, beamwidth and attenuation trade-offs. cite turn0search26

Note: Loss-budget values identified as estimates are engineering assumptions for this academic design and should be replaced by measured/manufacturer data in a production system.